EBeam_Javery – Imbalanced Mach-Zehnder interferometers, TE 1550 nm (final)

Designer: Javery (edX) · GitHub: ChenJavery · File: `submissions/EBeam_Javery.oas` · openEBL-2026-10 · SiEPIC EBeam PDK, KLayout + SiEPIC-Tools 0.5.31

1. Design concept

> **Question:** what is the group index n_g of the fabricated 500 × 220 nm strip waveguide, and does it match the model (n_g = 4.2036 from the course compact model; 4.24 from my MODE simulation)?
> **Method:** an imbalanced MZI produces fringes with FSR = λ^2 / (n_g·ΔL). Measuring FSR for several ΔL and fitting FSR against 1/ΔL gives n_g from the slope, independent of the grating-coupler loss.

Design decisions and why:

- **Five ΔL values (51, 101, 151, 201, 251 µm)** → FSR 11.2, 5.66, 3.78, 2.84, 2.28 nm. All inside the 1–20 nm window, at least 9 fringes in the 1499–1601 nm measurement sweep, and ≥ 110 points per fringe at its 20.4 pm step (PDK DFT: 5000 points). Five points make the linear fit and its residuals meaningful.
- **MZI6 = copy of MZI2 (ΔL 101 µm)** placed in the other row: the difference MZI6 − MZI2 measures on-chip (local) variation of n_g and fabrication repeatability.
- **Two GC loopbacks (LB1, LB2)**: they record the GC envelope used to normalise each MZI spectrum, and their difference shows GC-to-GC variation, i.e. the uncertainty of the de-embedding.
- **Identical building blocks**: every MZI uses the same PDK Y-branch, 500 nm strip waveguide and 5 µm bends; only the long-arm detour changes, so ΔL is the single design variable (ΔL in every cell name was checked against the routed length).
- **Balanced loss**: both arms are in the same waveguide, so arm imbalance is only the propagation loss of ΔL (≤ 0.075 dB) → propagation-loss limit on the extinction ratio > 47 dB (in practice the Y-branch split-ratio error sets the measured ER).

2. Layout and measurement labels

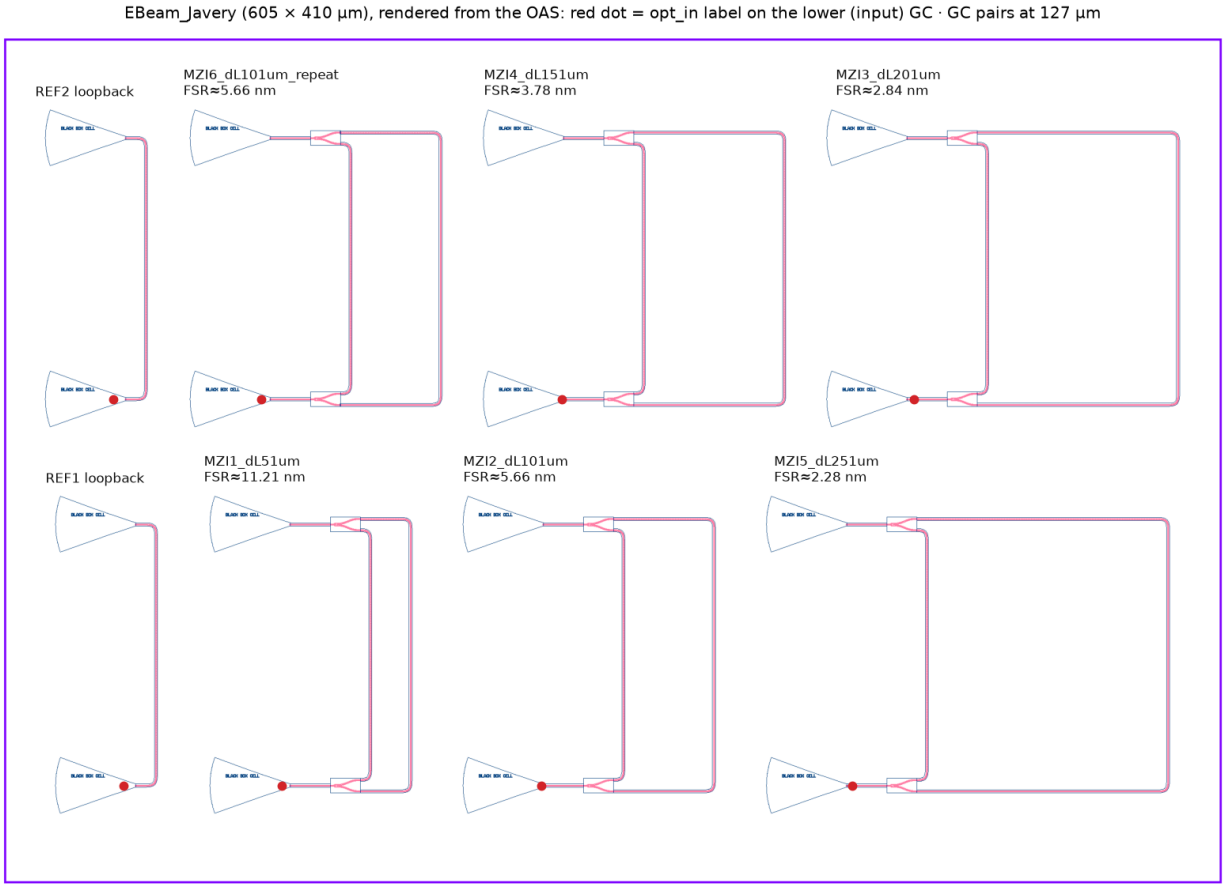


Fig. 1 – EBeam_Javery, 605 × 410 µm. Each circuit is a named sub-cell whose origin is its input GC; Rendered by KLayout from the submitted OAS (Si 1/0 in pink; GCs shown by their PDK outline — black-box cells replaced at fabrication). Red dot = opt_in label (2nd GC from top = lower GC); GC pairs at 127 µm pitch.

Cell	opt_in label	Input GC (µm)	ΔL (µm)	FSR (nm)	FSR at corners (nm)	Purpose
REF1_GC_Loopback_TE1550	opt_in_TE_1550_device_Javery_LB1	(65, 47)	—	—	—	GC envelope reference (de-embedding)
REF2_GC_Loopback_TE1550	opt_in_TE_1550_device_Javery_LB2	(60, 235)	—	—	—	GC envelope reference (de-embedding)
MZI1_dL51um_TE1550	opt_in_TE_1550_device_Javery_MZI1_dL51um	(142, 47)	51	11.207	11.02–11.20	FSR vs ΔL point
MZI2_dL101um_TE1550	opt_in_TE_1550_device_Javery_MZI2_dL101um	(268, 47)	101	5.659	5.56–5.65	FSR vs ΔL point
MZI6_dL101um_repeat_TE1550	opt_in_TE_1550_device_Javery_MZI6_dL101um	(132, 235)	101	5.659	5.56–5.65	repeat of MZI2: on-chip repeatability
MZI4_dL151um_TE1550	opt_in_TE_1550_device_Javery_MZI4_dL151um	(278, 235)	151	3.785	3.72–3.78	FSR vs ΔL point
MZI3_dL201um_TE1550	opt_in_TE_1550_device_Javery_MZI3_dL201um	(449, 235)	201	2.843	2.79–2.84	FSR vs ΔL point
MZI5_dL251um_TE1550	opt_in_TE_1550_device_Javery_MZI5_dL251um	(419, 47)	251	2.277	2.24–2.27	FSR vs ΔL point

Labels follow `opt_in_TE_1550_device_<username>_<device>`; the device part contains ΔL so measured data files are self-describing. Human-readable notes (e.g. "MZI3: dL=201 um") are on the Text layer next to each circuit. Input = lower GC, output = upper GC.

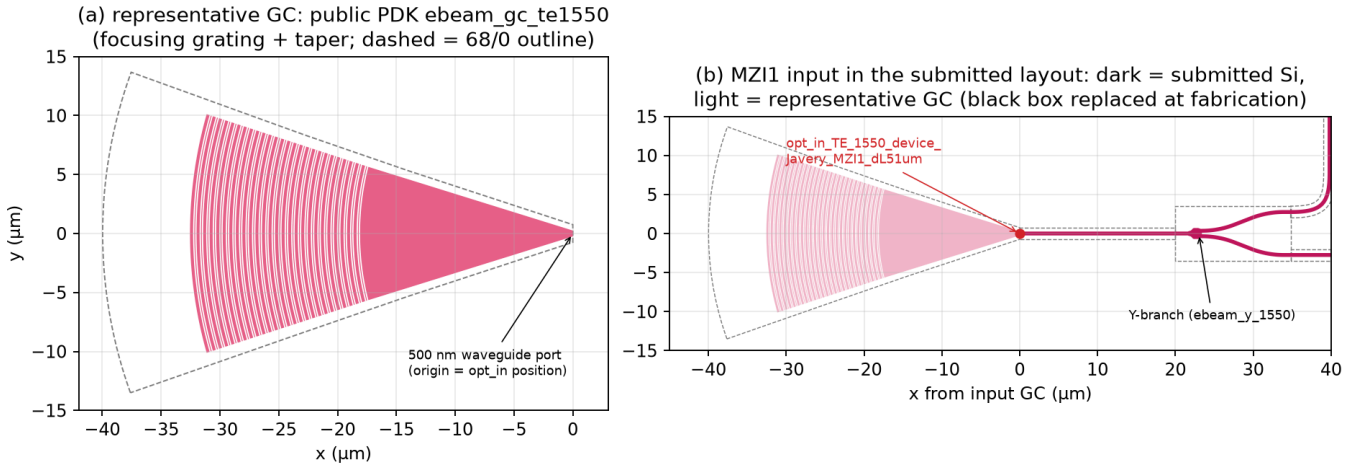


Fig. 1b – Close-up. The submitted GC cell (GC_TE_1550_8degOxide_BB) is a black box: only its outline and a 0.4 µm Si port are in the layout, and the foundry inserts the grating at fabrication (the cell must not be edited). For illustration the public PDK cell ebeam_gc_te1550 (identical outline) is drawn at the same origin: 52 curved grating teeth (x ≈ −32.5 … −17.5 µm) feeding an ≈18 µm taper into the 500 nm waveguide, followed by the Y-branch 20 µm downstream.

3. Expected response and robustness

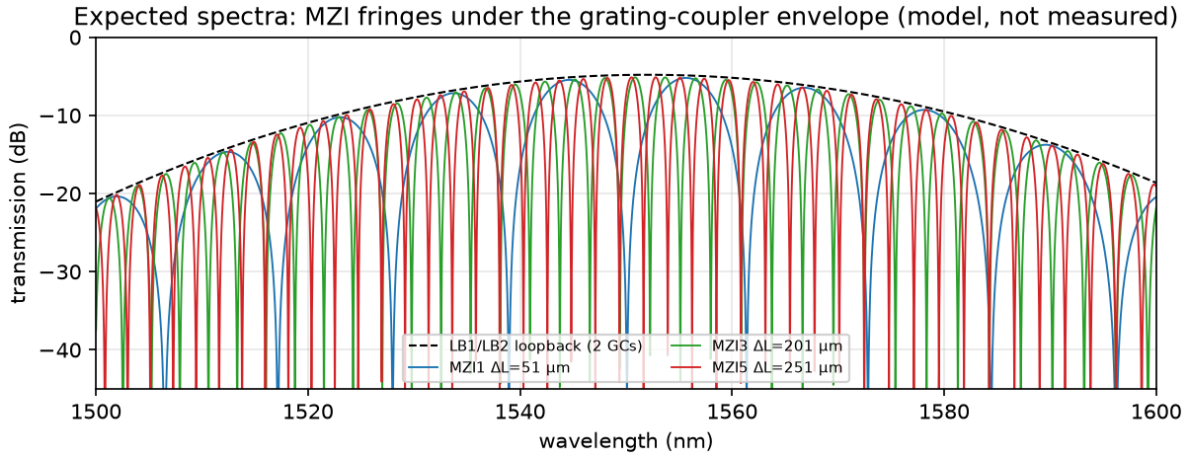


Fig. 2 – Expected transmission: analytic MZI (course compact model $n_{eff}(\lambda)$) under a GC envelope matched to the instructor's loopback simulation (2.4 dB per GC at 1552 nm) and 0.16 dB excess loss per Y-branch (my 2.7.5 varFDTD result).

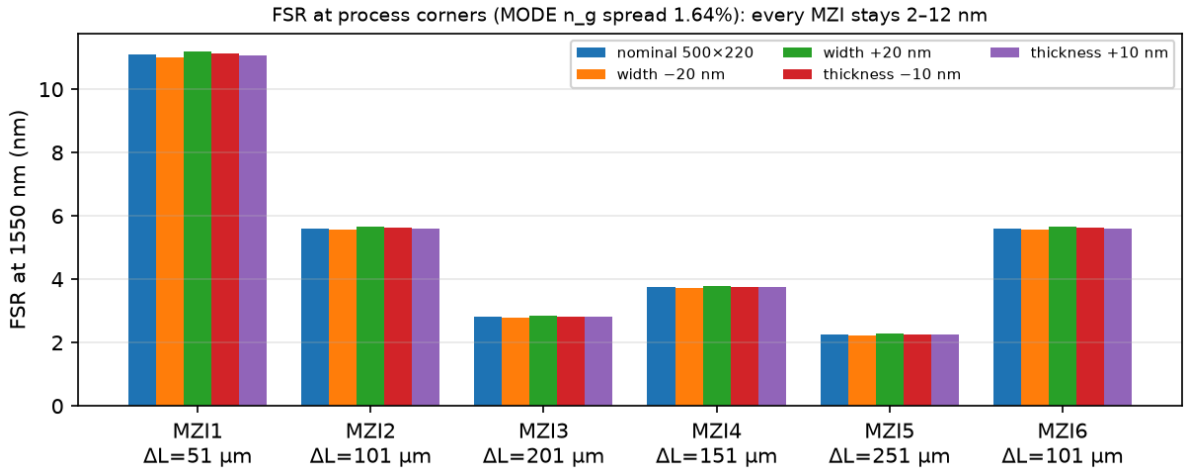


Fig. 3 – FSR at process corners using n_g from my MODE FDE sweep (unit 6.3): width ±20 nm, thickness ±10 nm (assumed tolerances, not foundry data).

MZI	Peak insertion loss (dB)	Arm imbalance (dB)	ER limit (dB)
MZI1	5.13	0.015	61.1
MZI2	5.14	0.03	55.2

MZI3	5.15	0.06	49.2
MZI4	5.14	0.045	51.7
MZI5	5.16	0.075	47.3
MZI6	5.14	0.03	55.2

Loss budget assumptions: 2.4 dB per GC (instructor's simulation of LB1/LB2, Sec. 5), 0.16 dB per Y-branch, 3 dB/cm waveguide. Cross-talk: no two Si edges are closer than 3 μm outside the Y-branch and GC cells; the evanescent field decays by e at every 0.13 μm (γ = 7.97 /μm), so coupling at ≥ 3 μm is negligible.

4. Measurement and analysis plan

1. Automated measurement of all 8 opt_in devices: TE, 1499–1601 nm, 5000 points (PDK DFT settings).
2. Subtract the loopback spectrum (LB1, LB2 average; their difference = de-embedding uncertainty) to remove the GC envelope.
3. Find fringe minima, compute FSR(λ) for each MZI; $n_g = \lambda^2 / (\mathrm{FSR} \cdot \Delta L)$; fit FSR vs $1/\Delta L$ across the five ΔL.
4. Compare with simulation (4.20–4.24); compare MZI2 vs MZI6 for repeatability; fit the full transfer function (n_{eff}, n_g, ER) as in unit 7.4.

5. Design prediction vs instructor circuit simulation

After the final upload (PR #747) the instructor ran a SiEPIC circuit simulation of all 8 measurement labels (2026-10-14). The plots (Fig. 5, used with the instructor's permission) were digitised to read the FSR at 1550 nm from the spacing of the transmission nulls (x calibrated to the 1499–1601 nm DFT sweep; ≈0.13 nm per pixel, averaged over 9–45 nulls).

MZI	ΔL (μm)	FSR design (nm)	FSR simulated (nm)	Difference	n_g from simulation
MZI1	51	11.207	11.248	+0.37%	4.188
MZI2	101	5.659	5.68	+0.37%	4.188
MZI6	101	5.659	5.68	+0.37%	4.188
MZI4	151	3.785	3.79	+0.13%	4.198
MZI3	201	2.843	2.843	+0.00%	4.204
MZI5	251	2.277	2.283	+0.26%	4.193

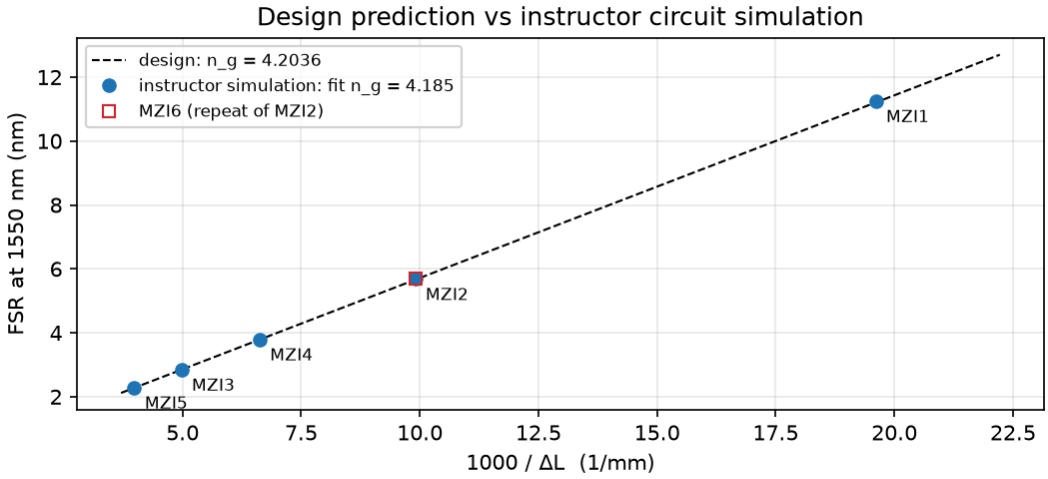


Fig. 4 – FSR vs 1/ΔL. The straight-line fit of the simulated FSRs gives n_g = 4.185 with intercept -0.009 nm (≈ 0, i.e. self-consistent), 0.4% from the design value 4.2036. MZI6 (repeat) is identical to MZI2, as expected for a simulation without fabrication variation.

- **Agreement:** every simulated FSR is within 0.4% of the design prediction; all MZIs show clean fringes with nulls below −45 dB.
- **GC loss:** the simulated loopbacks LB1/LB2 peak at ≈ −4.8 dB near 1551.7 nm, i.e. ≈ 2.4 dB per GC; Sec. 3 now uses this value (the earlier version of this report assumed a pessimistic 5 dB).
- **What measurement adds:** the simulation uses nominal geometry, so MZI2 = MZI6 exactly; on the chip their difference and the deviation of n_g from 4.19 will measure fabrication variation.

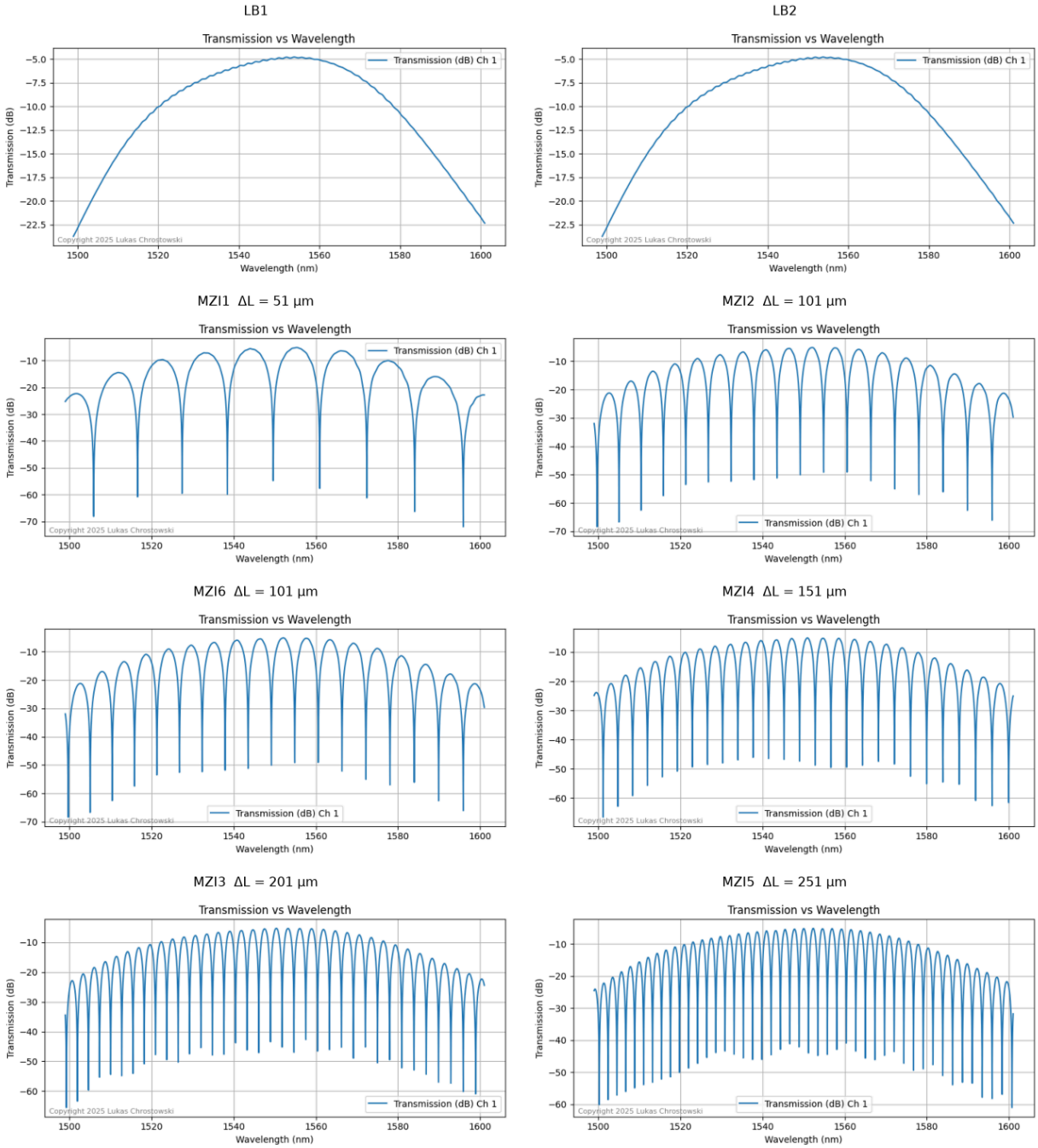


Fig. 5 – Instructor's circuit simulation of every opt_in device (L. Chrostowski, SiEPIC circuit simulation on PR #747, used with permission).

6. Design review checklist (course list) and verification

Checklist item	Status	Evidence
Adequate test structures / parameter variation (FSR 1–20 nm)	**PASS**	6 MZIs, 5 distinct ΔL (51–251 µm) → FSR 11.2–2.3 nm, + 1 repeat
De-embedding structures	**PASS**	two GC loopbacks (LB1 lower row, LB2 upper row)
DRC; all features ≥ 60 nm	**PASS (local)**	PDK DRC 0 items; extra 60 nm width/space check 0 (GC black-box interior not visible)
Avoids minimum-size features	**PASS**	only PDK cells + 500 nm strip waveguides; smallest drawn Si features are inside the PDK Y-branch / GC
Works under process variation	**PASS (model)**	MODE n_g spread 1.64% for width ±20 nm / thickness ±10 nm → every FSR stays in 2–12 nm; n_g extraction uses FSR·ΔL, so it is self-consistent
All waveguides connected	**PASS**	SiEPIC functional check 0 errors (pins / DevRec)
Paths converted, adequate radius	**PASS**	PDK waveguides, 5 µm Bezier bends, no sharp corners
Low loss / no cross-talk	**PASS (estimate)**	no Si spacing < 3 µm outside Y/GC; circuits > 20 µm apart; loss budget table
Correct layers	**PASS**	Si 1/0, PinRec 1/10, DevRec 68/0, Text 10/0, FloorPlan 99/0
Inside floor-plan	**PASS**	605 × 410 µm; Si extent x 60–584, y 44–365 µm
Space-efficient	**PASS**	8 circuits in two rows, evenly distributed (equal side margins and gaps, rows centred vertically); white space is only inside the MZI loops
GCs: 127 µm pitch, correct direction	**PASS**	GC_TE_1550_8degOxide_BB, R0, pairs at 127 µm
Label on 2nd GC from top	**PASS**	each label sits exactly on the lower (input) GC origin
Unique measurement labels	**PASS**	8 unique opt_in labels, each containing ΔL
File and cell names	**PASS**	EBeam_Javery.oas, top cell EBeam_Javery (edX username Javery)

Verification of the saved OAS (SHA-256 44cbd73f60b7e7d5…): SiEPIC functional/DFT 0 errors, PDK manufacturing DRC 0, 60 nm check 0. Local checks; the GitHub Actions verification and the merged EBeam layout are checked after upload.

7. Changes since the draft (response to review)

Review point	Draft	Final
Too much empty area	LB + 3 MZIs in the lower half	8 circuits evenly distributed over the full area (added MZI4, MZI5, MZI6, LB2)
Design concept unclear	three MZIs, purpose not stated	explicit goal (n_g extraction), five-point ΔL series, repeat device and two references with stated purpose
Labels unclear	generic labels on the upper GC	label on the 2nd GC from top (input), descriptive names with ΔL, named sub-cells, readable notes
Report not detailed	one page	concept, label table, expected spectra, corners, loss / cross-talk, analysis plan, full checklist
Process variation / loss not addressed	—	MODE corner analysis and loss budget (Sec. 3)
Prediction not checked	—	design FSRs confirmed by the instructor's circuit simulation within 0.4%; GC loss updated (Sec. 5)